School of Computer Science & IT, DAVV, Indore
End Semester Examination, July – Dec 2023
BCA(Sec_B) – 3rd Semester
CS-2133: Statistics and Probability

Time: 2.5 hrs **Max Marks: 60**

Note: Attempt **any five** questions. All carry equal marks. Answer the question at one place.

Q. No.	Question	Marks	CO
Q.1(a)	What do you mean by the measures of central tendency? Differentiate among Mean, Mode and Median in detail with examples	6	1
(b)	Find out the measurement of dispersion of the following series Class interval: 10-15, 15-20, 20-25, 25-30, 30-35, 35-40, 40-45 Frequency: 4, 8, 16, 12, 9, 5, 2 Find out (i) Quartile Q1 and Q3 (ii) Mean Deviation (iii) Standard Deviation	6	1
Q.2(a)	If P(A) = 0.65 , P(B) = 0.75 find the value of (i) $P(A \cup B)$ (ii) $P(A^C \cup B^C)$ (iii) $P(A^C \cap B^C)$ (iv) $P(A \cap B^C) - P(A^C \cap B)$ (v) $P(A \cap B) + P(A^C \cap B^C)$	6	2
(b)	Define the Bay's Theorem. There are three different boxes containing different color of balls. If box one contains 4 white ball, 5 red balls and 6 green balls, box two contains 5 white balls, 6 red balls and 4green ball, and box three contains 6 white balls, 4 red ball and 5 green balls. If two balls are drawn form a box chosen at random. These are found to be one white and one red. To find the probability that the balls so drawn came from second box.	6	2
Q.3(a)	Define Poisson Distribution. Write down the characteristics and application of Poisson Distribution.	6	3
(b)	If the probability of a bad reaction from a certain injection is 0.001, determine the chance that out of 4000 individuals more than two will get a bad reaction. Find out the mean, standard deviation, moment of coefficient of skewness and moment of coefficient of kurtosis.	6	3
Q.4(a)	Find a curve fitting of straight line for the following data X: 3, 7, 9, 6, 5, 10 Y: 13, 20, 22, 14, 16, 21	6	3

(b)	Find a curve fitting of second degree parabola to the following data. X: 0 1 2 3 4 Y: 1 1.5 1.3 2.5 5.3	6	3
Q.5(a)	Explain the correlation and regression. What are differences between correlation and regression?	6	4
(b)	Find out Spearman's Rank Correlation of the following data: X: 20 22 28 23 30 30 23 24 30 25 Y: 28 24 24 25 26 27 32 30 35 40	6	4
Q.6(a)	Define Hypothesis with example, and also Explain the characteristics of hypothesis.	6	5
(b)	Briefly describe the important parametric tests used in context of testing hypothesis.	6	5